

Bucket Foundation — Session Checkpoint v2

bkt-nuc · 2026-05-03 — adds quantum-plates batch + iterations

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This is the **second checkpoint of the day**. It supersedes the earlier 2026-05-03-session-checkpoint.md (still in checkpoints/ for history).

If you're recovering this session: read this file. Run `bd-remote list` and `git status` before changing anything.

TL;DR — what's ready, what needs review, what's pending

 **Ready for the web (you can pick from these now)**

Asset	Where	Notes
Canon globe	shipped at /canon	founder verdict: 7/10, satisfactory, keep as-is
Six DALL-E bobber concepts	gdrive manifesto-source/bobber-concept-art/2026-05-03/ (link)	review and pick 1–2 to drive the R3F build
	gdrive manifesto-source/quantum-	review and pick which go on /quantum

Asset	Where	Notes
Thirteen chalkboard math plates	plates/2026-05-03/ (link)	
Generator script	manifesto-source/ quantum-plates/ 2026-05-03/ generate_plates.py	every plate is a Python function, deterministic, modify and re-run to iterate

👁️ Needs founder review (in priority order)

1. **Pick winning bobber concept** (1–2 of the 6) → drives the R3F shader work
2. **Pick which plates ship on /quantum** (probably 4–6 of the 13)
3. **Confirm voice on the manifesto headline** (Decentralized research. The independent researcher succeeds. — yes/no/edit)
4. **Confirm bkt-6c5 scope** — globe stays at /canon, bobber lives at /quantum — update bead

🕒 Pending (waiting on you)

Bead	What	Blocking
bkt-ow0	iPad drawings + math upload	drop photos at ~/agfarms/bucket-foundation/manifesto-source/iPad-drawings-2026-05-03/; I'll mirror to gdrive and transcribe
bkt-6c5	R3F bobber-on-water build	needs winning concept + plate selection above

🔧 Technical debt logged this session (for a separate eai-bead)

The enterprise-ai/scripts/provision-instance.sh template has four bugs that were hot-patched on the live bucket-foundation instance but not fixed upstream:

1. IMAGE_TAG placeholder never substituted in statefulset manifest

- 2. Wrong image name `farmera/nucleus-orchestrator` (should be `farmera/nucleus`)
- 3. Naive `sed s/INSTANCE_ID/.../g` clobbers the `configmap` env-var **key**, leaving `INSTANCE_ID` unset → new instances return umbrella beads instead of their own scope
- 4. Ingress missing `spec.ingressClassName: traefik` (annotation-only is ignored by traefik 3.3+ in newer k3s)

File against `enterprise-ai` next time you're working on platform.

1. The thirteen quantum plates — full inventory

Real math, parametrized exactly, rendered with `matplotlib` in a chalkboard register (deep slate background `#1d2d2c`, off-white chalk strokes `#f4ead5`, antique gold accent `#B8861E`, slight hand-drawn jitter via `path.sketch = (1.5, 80, 1.5)`, serif italic labels). No AI image generation — every plate is a Python function with a parametric formula. The script IS the recipe.

The pairing structure

The plates pair across abstraction levels — this is how I'd lay them out on the page:

Pair	Theme	Plates
A. rotation → time	Euler's identity becomes a coil when time is added	01, 02
B. growth	the spiral that is invariant under scaling	03
C. paths on a curved surface	optimal vs. practical	04, 05
D. topology	torus, seed, iterated seed, the simplest knot	06, 07, 08, 13
E. dynamics	the bobber's heart, the bobber's wave	09, 10
F. discrete bridge		11, 12

Pair	Theme	Plates
	logic atom, the universal flattening map	

Status of each plate (post-iteration)

#	Plate	Status	Notes
01	Euler's identity	✓ ship-ready	the chalkboard wobble feels professorial
02	Helix $\gamma(t)$	✓ ship-ready	clean coil, shadow on $z=0$
03	Logarithmic spiral	✓ ship-ready	iterated to remove jitter artifact, gold tangent visible
04	Great-circle geodesic	✓ ship-ready	sphere now fills frame after tight_3d fix
05	Loxodrome	✓ ship-ready	same tight_3d fix
06	Torus	✓ ship-ready	gold flow line traces the toroidal field
07	Vesica piscis	✓ ship-ready	lens shaded gold, vertices labeled (0, $\pm\sqrt{3}/2$)
08	Flower of life	✓ ship-ready	19-circle hex iteration, gold center highlighted
09	Phase portrait	✓ ship-ready	iterated to use normalized phase space ($x, \dot{x}/\omega$) — now true honest spirals
10		✓ ship-ready	

#	Plate	Status	Notes
	Radial wave pulse		iterated to fewer levels (3 crests + 3 troughs), reads cleaner
11	NAND gate	✓ ship-ready	gate + truth table, gold accent on the 1s
12	Stereographic projection	✓ ship-ready	added in this batch, sphere is now round (equal-axis-lim fix)
13	Trefoil knot	✓ ship-ready	added + redone as 2D knot diagram with proper over/under crossings (textbook register) — major upgrade from the original 3D version

My pick for /quantum page hero set

If I had to pick four for the page right now, in order:

1. **06 torus** — the load-bearing geometry, the central object of every layer-4 metaphor (heart EM, magnetosphere, fusion plasma)
2. **09 phase portrait** — the bobber's heart, the dynamical fingerprint of every oscillator
3. **07 vesica piscis** — the two-circle seed, mapping directly to your “central + decentral nodes” architecture
4. **10 radial wave pulse** — the bobber's outgoing field, $\psi(r,t) = \sin(kr - \omega t)/\sqrt{r}$

01 (Euler) is the natural opening above this set. 11 (NAND) is the natural closer. That's a six-plate scroll.

13 (trefoil) and 08 (flower) are good supplementary pull-quotes. 04, 05, 12 are good for the deeper “geometry on surfaces” appendix.

2. The six bobber concepts (DALL-E 3, HD)

Generated in this session, mirrored to gdrive.

#	Slug	Costume	Verdict
01	bobber-on-still-water-photoreal	literal bobber, gold concentric rings	photographic register; reads “Apple TV+ logo”; portal-circle frame is gimmicky
02	torus-spinning-around-sphere-architectural	60s physics-textbook plate, magnetic field topology	DALL-E added a painterly frame and dark void; doesn't match bone palette
03	vesica-piscis-flower-of-life-pulsating	sacred geometry, parchment palette	static — it's the seed but not the engine
04	quantum-particle-wave-duality	sphere + sinusoidal wave + torus field	most legible — only frame showing all four mechanical elements
05	earth-as-bobber-cosmic-water	planet-as-bobber, starfield, ripples	most cinematic; earth is too literal/illustrated
06	heart-as-torus-electromagnetic-field	toroidal heart EM field	physics correct (HeartMath, etc.); flirts with

#	Slug	Costume	Verdict
			woo for public face

My pick: #04 (particle-wave duality) is the strongest hero-frame candidate. **#01 (bobber-photoreal)** is best for the founders-narrative track. **#03 (vesica/flower)** belongs on the geometry appendix page alongside the chalkboard plates.

3. Manifesto voice — the four-layer frame

Built this session, locked:

Layer	What	One-line
1. architecture	how Bucket works	<i>decentralized research</i>
2. human	who Bucket serves	<i>the independent researcher succeeds</i>
3. civilizational	what Bucket unlocks	<i>fastest path to innovation</i>
4. cosmological	why this geometry	<i>awareness × direction · circle × line · pulsating — the math plates and (pending) iPad drawings live here</i>

Current spine

Six words: > Decentralized research. The independent researcher succeeds.

One sentence: > Bucket is the rail where reading is free, citing pays the author, and > the work compounds forever.

Voice rule the founder added this session

Stop using the negation stack (“not a. not a. then it is X”). Be direct. Affirm without negating first. This rule overrides the prologue pattern from papers/book/.

Architecture in plain words (from founder, condensed)

1. **Read free. Train free.** No paywall on data, papers, datasets, AI training sets.
2. **Cite paid.** When output goes commercial — paper, model, product, public chain — you owe the source. Citation = payment, x402, no Elsevier.
3. **Equity in the citation graph.** Author isn't paid once; they hold a position in every downstream graph their work touches. Forever.
4. **Citations as options.** Capped supply, bid markets, funded bounties, replication contracts.
5. **Two kinds of nodes, both legal.** Centralized Bucket sub-nodes (funded, branded) and decentralized peer nodes (local AI, ZK-proven labs). Cite from either rail.

What Bucket actually is (under the manifesto)

The citation rail for the AI commercial era. Bucket sits underneath every commercial AI output and asks: who did you read, and did they get paid?

4. Repo state at checkpoint v2

- **Repo:** gianyrox/bucket-foundation (public, MIT)
 - **Branch:** main
 - **Commits this session (in order):**
 - 63a9a4b — feat(site): scale canon globe to hero size
 - 4d8e443 — feat(site): canon globe visual polish (denser dots, beamed markers, layered halo, starfield)
 - 28d1c2d — docs(checkpoints): v1 session checkpoint
 - 6530232 — docs(manifesto): bobber concept art INDEX
 - 27f5dcc — docs(manifesto): chalkboard quantum plates (initial batch of 11)
 - *this commit* — docs(manifesto): quantum plates iterated (4 fixes + 2 new) + checkpoint v2
 - **Auto-deploy:** Vercel from main
 - **Working tree at save:** clean
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5. Next session — pick up here

You (founder), in any order: 1. Drop iPad photos at `~/agfarms/bucket-foundation/manifesto-source/iPad-drawings-2026-05-03/` to close `bkt-ow0`. 2. Pick the winning bobber concept (#04 is my recommendation). 3. Pick the plate set for `/quantum` (my recommendation: 01, 06, 07, 09, 10, 11 — six-plate scroll). 4. Confirm or edit the manifesto headline.

Me (next agent), once you've decided: 1. Update `bkt-6c5` description: globe stays at `/canon`, bobber + plates live at `/quantum`. 2. Scaffold `src/components/quantum-bobber/` (the R3F shader work, plan in v1 checkpoint). 3. Scaffold `src/app/quantum/page.tsx` — pulls plate PNGs from the static folder, weaves them into a scroll story between the bobber hero and the closing line. 4. Mirror plate PNGs from gdrive into `/public/quantum-plates/` for the build (gitignored locally, fetched at deploy time, OR committed to `public/` since they're small). 5. Once iPad math is up, transcribe relevant equations and consider a `GEOMETRY.md` doc.

Generated by bd-nuc · 2026-05-03 (v2 supersedes v1).